GOVERNMENT OF PUNJAB

OFFICIAL e-PROCUREMENT TENDER NOTIFICATION · River Basin Hydro-Meteorological Telemetry Sensor Network & GIS Flood Simulation

DETAILED TECHNICAL SPECIFICATIONS & SCOPE DOCUMENT

Tender Reference ID: 2026_PUN_6047755_24 Portal: https://tender.punjab.gov.in

Department / Authority: WATER RESOURCES & IRRIGATION DEPARTMENT

Published Date: 21 Oct 2026 | Online Bid Closing: 18 Nov 2026, 15:00 hrs

Tender Type: Open Competitive Global / National e-Bidding | Form of Contract: Turnkey Item Rate

EMD Value: 2% of Bid Value via e-Gateway | Tender Document Fee: INR 10,000 + 18% GST (Non-refundable)

1. EXECUTIVE SUMMARY & INVITATION TO BID

1.1 The Government of Punjab, acting through WATER RESOURCES & IRRIGATION DEPARTMENT, invites eligible and qualified agencies to submit technical and financial proposals online through the official state e-procurement portal for:

"Water Resources-Based River Basin Hydro-Meteorological Sensor Network & GIS Flood Simulation"

1.2 Brief Purpose: Turnkey design, supply, installation, testing and commissioning of automated hydro-meteorological telemetry stations, radar water level sensors, automated rain gauges (ARG), and real-time SCADA IoT gateway infrastructure. Coupled with 1D/2D hydrodynamic flood simulation models and web-GIS early warning emergency decision dashboards.

1.3 Bidders are advised to thoroughly inspect the technical guidelines, bill of quantities (BOQ), minimum eligibility criteria, and general condition clauses before uploading their signed and sealed proposals.

2. CRITICAL DATES SHEET & PROCUREMENT TIMELINES

Milestone Event Description	Official Schedule / Timestamp
Official Portal Publication Date	21 Oct 2026
Document Download Start Date	21 Oct 2026, 10:00 hrs
Pre-Bid Query Submission Deadline	Within 7 Days of Portal Publication
Pre-Bid Clarification Meeting (Hybrid/VC)	10th Day of Publication at 11:30 hrs
Bid Submission Start Date	21 Oct 2026, 14:00 hrs
Online Bid Submission Due Date & Time	18 Nov 2026, 15:00 hrs
Technical Bid Opening Date & Time	Next Working Day following Bid Closing at 15:30 hrs
Financial Bid (BOQ) Opening Date	To be notified online to technically qualified bidders

3. COMPREHENSIVE TECHNICAL SCOPE & ENGINEERING METHODOLOGY

3.1 Overview of Engineering Operations:
The selected agency shall be responsible for turnkey end-to-end execution of all technical requirements described herein. All work must adhere strictly to National Geospatial Policy guidelines, Survey of India precision benchmarks, and relevant state standards.

3.2 Key Technical Components:
? Phase I ? Continuous Operating Reference Station (CORS) & DGPS Network: Establishment of primary Ground Control Points (GCPs) with Dual Frequency GNSS receivers linked to the state CORS network, achieving sub-2cm absolute spatial positioning accuracy.
? Phase II ? High-Precision Aerial / Terrestrial Sensor Acquisition: Deployment of calibrated multi-rotor/fixed-wing survey drones or Mobile Mapping Systems (MMS) equipped with high-resolution digital metric cameras and LiDAR sensors capable of delivering 5 cm Ground Sample Distance (GSD) orthophoto mosaics.
? Phase III ? Vector Extraction & Feature Classification: Digitization of building footprints, road alignments, parcel boundaries, and municipal infrastructure with structured attribute schemas conforming to national spatial data exchange standards.
? Phase IV ? Comprehensive Field Verification & Attribute Geo-tagging: Door-to-door verification using custom mobile GIS applications with automated GPS coordinates, photo capture, and offline cache synchronization.
? Phase V ? Enterprise Geodatabase & Decision Support Web GIS: Development and hosting of an enterprise PostgreSQL / PostGIS spatial database with GeoServer map services, OGC WMS/WFS interfaces, and role-based access control for department administrators.

3.3 Statutory Standards, Geodetic Accuracy & Delivery Format Requirements

? Coordinate Reference System: WGS84 Datum / UTM Projection (conforming to state UTM Zone).
? Horizontal Positional Accuracy: RMSE <= 5.0 cm on well-defined ground features.
? Vertical Precision: RMSE <= 10.0 cm for Digital Surface Model (DSM) and Digital Terrain Model (DTM).
? Orthomosaic Delivery: Uncompressed GeoTIFF and ECW format with complete pyramidal tiles.
? Vector Delivery: ESRI File Geodatabase (.gdb) and OGC GeoPackage (.gpkg) with topological validation.
? Attribute Data: Linked SQLite / PostgreSQL relational tables with audited integrity constraints.
? Cloud Deployment: High-availability tier with 99.9% uptime SLA and periodic daily automated snapshots.

3.4 Quality Control, Assurance & Third-Party Audit

The department reserves the right to appoint a Third-Party Quality Inspection Agency (TPIA) or Survey of India / State Remote Sensing Centre to validate 10% randomly selected sample plots across all deliverables. Any deliverable failing accuracy tolerances shall be resurveyed by the vendor at no additional cost to the department.

4. MINIMUM BIDDER QUALIFICATION & TECHNICAL ELIGIBILITY CRITERIA

Bidders must meet all of the following mandatory pre-qualification conditions as on the date of bid submission. Consortiums/Joint Ventures (JV) are permitted as per the conditions stated in Section 5.

Criteria Category	Mandatory Qualification Threshold	Documentary Proof Required
Annual Financial Turnover	Average annual turnover of at least INR 22.00 Crores over the last 3 financial years (FY 2023-24, 2024-25, 2025-26).	Audited Balance Sheets & CA Certificate with UDIN
Past Experience	Proven track record in river basin telemetry, dam SCADA, or National Hydrology Project (NHP) contracts.	Client Completion Certificates & Work Orders
Quality Certifications	Valid ISO 9001:2015, ISO 27001 (Information Security), and minimum CMMI Level 3 (Services/Dev).	Valid Certificate Copies from Accredited Registrar
Legal Status	Registered Company under Indian Companies Act (or LLP) in operation for at least 5 years.	Certificate of Incorporation & GST Registration
Solvency Status	Bank Solvency Certificate for an amount not less than INR 8.80 Crores from a Scheduled Commercial Bank.	Original Bank Solvency Certificate in prescribed format
Debarment Undertaking	The bidder must not have been blacklisted or debarred by any Central/State Govt/PSU as on bid date.	Notarized Affidavit on INR 100 Stamp Paper

4.2 Mandatory Key Technical Personnel to be Deployed on Project

Designation / Role	Minimum Academic Qualification	Min Relevant Experience
Project Director / Team Leader	M.Tech / M.Sc Remote Sensing / GIS / Civil	12+ Years Turnkey Project Leadership
Senior GIS Database Architect	B.Tech / MCA with PostgreSQL/PostGIS / Oracle	8+ Years Spatial DB Architecture
Photogrammetry / LiDAR Specialist	M.Tech / B.Tech Geoinformatics / Photogrammetry	8+ Years Aerial Sensor Processing
Field Survey & QA Lead Engineer	Degree/Diploma Civil + DGCA Drone Pilot Cert	6+ Years DGPS & Ground Truthing
Full-Stack WebGIS Developer	B.Tech Computer Science / IT (React / Python)	5+ Years Web Map Portal Dev

5. PROJECT DELIVERABLE MILESTONES & PROGRESS PAYMENT SCHEDULE

Milestone	Deliverable Description	Target Timeline	Payment (%)
M-1	Inception Report, Project Execution Plan & Statutory Clearances	T0 + 30 Days	10%
M-2	Establishment of DGPS Ground Control Network & Raw Sensor Acquisition	T0 + 90 Days	20%
M-3	Ortho-mosaic Processing, Feature Extraction & 50% Field Verification	T0 + 180 Days	25%
M-4	100% Field Survey Completion, UPIN Geo-tagging & Database Validation	T0 + 270 Days	25%
M-5	Web GIS Portal Deployment, User Acceptance Testing (UAT) & Training	T0 + 330 Days	10%
M-6	Final Handover, Final Documentation Sign-off & Commencement of AMC	T0 + 365 Days	10%

6. SERVICE LEVEL AGREEMENT (SLA), PENALTIES & PERFORMANCE GUARANTEE

? Performance Security (PBG): The successful bidder shall furnish a Performance Bank Guarantee equal to 5% of the total contract value within 14 days of the Letter of Award (LoA), valid until 60 days after completion of all contractual obligations.
? Delay Liquidated Damages: If the contractor fails to deliver milestones within agreed timelines, liquidated damages of 0.5% per week of delay (or fraction thereof) shall be deducted, subject to a maximum cap of 10% of total contract value.
? Defect Liability & Maintenance: 12 months comprehensive warranty from the date of final operational sign-off.
? Data Security & Ownership: All raw aerial imagery, point clouds, GIS geodatabases, and field registers shall be the sole and exclusive property of the Government of Punjab. The vendor shall not disclose or retain proprietary government data.

ISSUED FOR AND ON BEHALF OF:

Government of Punjab

WATER RESOURCES & IRRIGATION DEPARTMENT

[Digitally Signed & Validated on e-Procurement Portal]

Competent Tender Inviting Authority (TIA)